
Chen-Li 4.7

2026-08-10

Danus arXiv DOI

J. Chen and J. Li, *Mean Curvature Flow of Surface in 4-Manifolds*, Adv. Math. 163 (2001), 287-309, Theorem 4.7

Chen-Li 4.7

1. Wang (2001) Kähler-Einstein Li-Yang (2014) Zhang
(2015) Qiu-Sun (2019/2020) Kähler

2. Chen-Li (2002) Remark 6.7 “
”

3. Han-Li Han-Li-Sun Li-Sun

4. Chen-Li (2004) Neves (2007) Maslov

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Wang “ ” Chen-Li (2002)

1.

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- Chen-Li 4.7
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- A arXiv
- B DOI

$$\sup_{\Sigma_t} |A|^2 \leq \frac{C}{T-t}, \qquad t < T.$$

2.

			4.7 /
2000	Chen-Tian, <i>Moving Symplectic Curves in Kähler-Einstein Surfaces</i> [R1]	Theorem 2.4 Kähler Theorem 2.5	A
2001	Wang, <i>Mean Curvature Flow of Surfaces in Einstein Four-Manifolds</i> [R3]	Theorem A Kähler-Einstein Kähler Huisken	Chen-Li A
2002	Chen-Li, <i>Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces</i> [R4]	Theorems 1.2, 4.3, 5.1, 6.6 Remark 6.7	“ ” A
2014	Li-Yang, <i>Symplectic Mean Curvature Flows in Kähler Surfaces with Positive Holomorphic Sectional Curvatures</i> [R12]	2 Kähler Theorem 1.3 Chen-Li/Wang	Kähler-Einstein A
2014	Li-Yang, <i>Generalized Symplectic Mean Curvature Flows in Almost Einstein Surfaces</i> [R13]		MCF B
2015	Zhang, <i>Remarks on Symplectic Mean Curvature Flows...</i> [R14]	Li-Yang Kähler Theorem 3.2 /	arXiv [R12] A

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2019	Qiu-Sun, <i>Mean Curvature Flow of Surfaces in a Hyperkähler 4-Manifold</i> [R15]	Theorem 1.2 Theorem 1.3	White A
2026	Chen-Han-Li-Sun, <i>Tangent Flows of Symplectic Mean Curvature Flows</i> [R18]	Type I*	“ ” “ ” B

3.

			/
2009	Han-Li, <i>Singularities of Symplectic and Lagrangian Mean Curvature Flows</i> [R5]	Theorem 2.4 $\mathbb{C}^2$ $-2 \leq K \leq 0$ Corollary 2.6	A
2009	Han-Li, <i>Translating Solitons to Symplectic and Lagrangian Mean Curvature Flows</i> [R6]	/	A
2011	Han-Li-Sun, <i>The Second Type Singularities...</i> [R8]	Theorems 1.1-1.4 Kähler	A
2024	Li-Sun, <i>Eternal Solutions to Almost Calibrated Lagrangian and Symplectic Mean Curvature Flows</i> [R16]	Kähler /	B

4.

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2010	Han-Li, <i>The Mean Curvature Flow Approach to the Symplectic Isotopy Problem</i> [R7]	Kähler-Einstein	A
2011	Han-Li, <i>Long Time Existence of the Symplectic Mean Curvature Flow</i> [R9]	Kähler	arXiv A
2013	Han-Li-Yang, <i>Symplectic Mean Curvature Flow in $\mathbb{CP}^2$</i> [R10]	$\mathbb{CP}^2$	A/B
2014	Han-Li, <i>The Mean Curvature Flow Along the Kähler-Ricci Flow</i> [R11]	Kähler-Ricci MCF /	A
2024	Han-Li-Sun, <i>Gradient Flow for β-Symplectic Critical Surfaces</i> [R17]	β - / β -	gradient-flow MCF A

5.

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2004	Chen-Li, <i>Singularity of Mean Curvature Flow of Lagrangian Submanifolds</i> [L1]	Calabi-Yau Theorem 1.1 2 Corollary 6.7	A
2007	Neves, <i>Singularities of Lagrangian Mean Curvature Flow: Zero-Maslov Class Case</i> [L2]	Theorems A-B Maslov	Maslov “ ” A
2017	Sun-Yang, <i>Generalized Lagrangian Mean Curvature Flows in Almost Calabi-Yau Manifolds</i> [L3]	Neves LMCF Corollary 1.1	A

			/
2021	Lambert-Lotay-Schulze, <i>Ancient Solutions in Lagrangian Mean Curvature Flow</i> [L4]	neck blow-down Lawlor blow-down	A
2024	Lotay-Schulze-Székelyhidi, <i>Ancient Solutions and Translators of Lagrangian Mean Curvature Flow</i> [L5]	Maslov blow-down Brakke	/ A
2025	Li-Luo-Sun, <i>Type II Singularities of Lagrangian Mean Curvature Flow with Zero Maslov Class</i> [L6]	Maslov	A

6.

Kähler	$\cos \alpha > 0$		[R1], [R2], [R3]
Huisken	$1/\cos \alpha$	/	[R2], [R4], [R8]
	$ A ^2 \leq C/(T-t)$		[R4], [L1]
			[R15]
	$(T-t) \max A ^2 \rightarrow \infty$	/	[R5], [R6], [R8], [R16]
	$ A ^2, H ^2, \cos \alpha$		[R7], [R9], [R10], [R12], [R14]

7. 4.7

7.1

1. Wang Theorem A

Chen-Li Theorem 4.7

Kähler-Einstein

2. Chen-Li 2002, Remark 6.7

3. Qiu-Sun Theorem 1.3

Kähler

7.2

- “ ” 1
- “ ” “ ” White
- Chen-Li 2002 “ ” Wang
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7.3

Chen-Li/Wang

“ ”

White

/

Chen-Li 2002 “ ”

8.

2026-08-10

• Li-Sun 2024 [R16]

• Han-Li-Sun 2024 [R17] β -

• Li-Luo-Sun 2025 [L6] Maslov

• Chen-Han-Li-Sun 2026 [R18] MCF Type I*

arXiv:2309.16478 2025-10-13

9.

- [R1] J. Chen and G. Tian, "Moving Symplectic Curves in Kähler-Einstein Surfaces," *Acta Math. Sin. (Engl. Ser.)* 16 (2000), 541-548. DOI: [10.1007/s101140000075](https://doi.org/10.1007/s101140000075). _____.
- [R2] J. Chen and J. Li, "Mean Curvature Flow of Surface in 4-Manifolds," *Adv. Math.* 163 (2001), 287-309. DOI: [10.1006/aima.2001.2008](https://doi.org/10.1006/aima.2001.2008).
- [R3] M.-T. Wang, "Mean Curvature Flow of Surfaces in Einstein Four-Manifolds," *J. Differential Geom.* 57 (2001), 301-338. DOI: [10.4310/jdg/1090348113](https://doi.org/10.4310/jdg/1090348113). [arXiv:math/0110019](https://arxiv.org/abs/math/0110019).
- [R4] J. Chen and J. Li, "Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces," *arXiv:math/0208227* (2002). [arXiv](https://arxiv.org/abs/math/0208227).
- [R5] X. Han and J. Li, "Singularities of Symplectic and Lagrangian Mean Curvature Flows," *Front. Math. China* 4 (2009), 283-296. DOI: [10.1007/s11464-009-0018-4](https://doi.org/10.1007/s11464-009-0018-4). [arXiv:math/0611857](https://arxiv.org/abs/math/0611857).
- [R6] X. Han and J. Li, "Translating Solitons to Symplectic and Lagrangian Mean Curvature Flows," *Int. J. Math.* 20 (2009), 443-458. DOI: [10.1142/S0129167X09005352](https://doi.org/10.1142/S0129167X09005352). [arXiv:0711.4435](https://arxiv.org/abs/0711.4435).
- [R7] X. Han and J. Li, "The Mean Curvature Flow Approach to the Symplectic Isotopy Problem," *J. Eur. Math. Soc.* 12 (2010), 505-527. DOI: [10.4171/JEMS/207](https://doi.org/10.4171/JEMS/207).
- [R8] X. Han, J. Li and J. Sun, "The Second Type Singularities of Symplectic and Lagrangian Mean Curvature Flows," *Chinese Ann. Math. Ser. B* 32 (2011), 223-240. DOI: [10.1007/s11401-011-0635-6](https://doi.org/10.1007/s11401-011-0635-6).
- [R9] X. Han and J. Li, "Long Time Existence of the Symplectic Mean Curvature Flow," *arXiv:1107.0829* (2011). [arXiv](https://arxiv.org/abs/1107.0829).
- [R10] X. Han, J. Li and L. Yang, "Symplectic Mean Curvature Flow in $\mathbb{CP}^2$," *Calc. Var. Partial Differential Equations* 48 (2013), 111-129. DOI: [10.1007/s00526-012-0544-x](https://doi.org/10.1007/s00526-012-0544-x).
- [R11] X. Han and J. Li, "The Mean Curvature Flow Along the Kähler-Ricci Flow," *arXiv:1105.1200* (2011). [arXiv](https://arxiv.org/abs/1105.1200).
- [R12] J. Li and L. Yang, "Symplectic Mean Curvature Flows in Kähler Surfaces with Positive Holomorphic Sectional Curvatures," *Geom. Dedicata* 170 (2014), 63-69. DOI: [10.1007/s10711-013-9867-9](https://doi.org/10.1007/s10711-013-9867-9). [arXiv:1207.5253](https://arxiv.org/abs/1207.5253).
- [R13] J. Li and L. Yang, "Generalized Symplectic Mean Curvature Flows in Almost Einstein Surfaces," *Chinese Ann. Math. Ser. B* 35 (2014), 33-50. DOI: [10.1007/s11401-013-0817-5](https://doi.org/10.1007/s11401-013-0817-5).
- [R14] S. Zhang, "Remarks on Symplectic Mean Curvature Flows in Kähler Surfaces with Positive Holomorphic Sectional Curvatures," *arXiv:1507.06806* (2015). [arXiv](https://arxiv.org/abs/1507.06806).

[R15] L.-J. Qiu and J. Sun, “Mean Curvature Flow of Surfaces in a Hyperkähler 4-Manifold,” arXiv:1902.00645 (2019, revised 2020). [arXiv](#).

[R16] X. Li and J. Sun, “Eternal Solutions to Almost Calibrated Lagrangian and Symplectic Mean Curvature Flows,” *J. Math. Anal. Appl.* 531 (2024), Article 127804. DOI: [10.1016/j.jmaa.2023.127804](#).

[R17] X. Han, J. Li and J. Sun, “Gradient Flow for β -Symplectic Critical Surfaces,” *Ann. Inst. H. Poincaré C Anal. Non Linéaire* 41 (2024), 1083-1116. DOI: [10.4171/AIHPC/100](#). [_____](#).

[R18] J. Chen, X. Han, J. Li and J. Sun, “Tangent Flows of Symplectic Mean Curvature Flows,” *J. Math. Study* 59 (2026), 40-59. DOI: [10.4208/jms.v59n1.26.03](#). [_____](#).

[L1] J. Chen and J. Li, “Singularity of Mean Curvature Flow of Lagrangian Submanifolds,” *Invent. Math.* 156 (2004), 25-51. DOI: [10.1007/s00222-003-0332-5](#). [arXiv:math/0301281](#).

[L2] A. Neves, “Singularities of Lagrangian Mean Curvature Flow: Zero-Maslov Class Case,” *Invent. Math.* 168 (2007), 449-484. DOI: [10.1007/s00222-007-0036-3](#). [arXiv:math/0608399](#).

[L3] J. Sun and L. Yang, “Generalized Lagrangian Mean Curvature Flows in Almost Calabi-Yau Manifolds,” *J. Geom. Phys.* 117 (2017), 68-83. DOI: [10.1016/j.geomphys.2017.03.001](#). [arXiv:1307.7854](#).

[L4] B. Lambert, J. D. Lotay and F. Schulze, “Ancient Solutions in Lagrangian Mean Curvature Flow,” *Ann. Sc. Norm. Super. Pisa Cl. Sci.* 22 (2021), 1169-1205. DOI: [10.2422/2036-2145.201901_016](#). [arXiv:1901.05383](#).

[L5] J. D. Lotay, F. Schulze and G. Székelyhidi, “Ancient Solutions and Translators of Lagrangian Mean Curvature Flow,” *Publ. Math. IHÉS* 140 (2024), 1-35. DOI: [10.1007/s10240-023-00143-5](#). [arXiv:2204.13836](#).

[L6] X. Li, Y. Luo and J. Sun, “Type II Singularities of Lagrangian Mean Curvature Flow with Zero Maslov Class,” *J. Funct. Anal.* 289 (2025), Article 111032. DOI: [10.1016/j.jfa.2025.111032](#). [arXiv:2412.15880](#).

10.

Chen-Li	4.7	Wang 2001	Chen-Li 2002	Li-Yang 2014	Zhang 2015	Qiu-Sun
2019/2020						
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	4.7		“	”		
		/	Chen-Li 2002			

Kähler

Qiu-Sun